KOR STRUCTURAL — OPERATIONS

Backup & Virtualization Infrastructure Findings & Remediation Punch List

For review with T-Net (managed backup provider)

Issued: 2026-07-16 · **Rev:** 2026-07-20 **Prepared for:** Ian Lalonde, Operations Lead

Scope: Veeam B&R 12.0.0.1420 (KOR-BK01) · vCenter 192.168.1.9 · ESXi .10/.16 · UC3200 SAN · Synology repo · Cloud Connect (remote.t-net.ca) · KOR-FS01

Context. An investigation following the 2026-07-13 Data Deduplication rollout on KOR-FS01 surfaced a chain of unmonitored backup-infrastructure failures — the primary one, an expired vCenter certificate, predating and unrelated to the dedup work. All findings verified against live console, repository, and database evidence 2026-07-15 → 07-20.

A0 · Resolution status (2026-07-20) — read first.

- **FS01 is protected again.** New job `Kor-FS01` → Synology repository, **forward incremental**, completed 07-20: **5.79 TB at ~126 MB/s in ~14 h, 0 errors.** First FS01 restore point since 07-07 — a 13-day gap, closed.
- **The measurement that matters:** the same workload via the existing **reverse-incremental** job to E: ran at **~29 MB/s** and consumed **~23 TB** of repository before failing. Forward incremental on a right-sized target: **~4× faster, ~⅓ the disk.**
- Also done: vCenter certificate re-accepted; FS01 snapshot confirmed removed + consolidated; no stray hot-add disks; BK01 patched and rebooted; C: reclaimed (was 20 GB free).
- **Still open:** off-site copy of FS01 (needs new copy job + physical seed); Veeam upgrade; mount server for file-level restore; E: structurally over-committed (A10); **no restore has ever been tested (A11)**.

A. Findings

Timeline — how an 8-day silent outage unfolded READ FIRST

- **Jul 7 ~17:00** — last successful backups (proven by restore-point files on disk).
- **Jul 7/8** — vCenter TLS certificate expires. Every nightly job + 4-hourly discovery begins failing. **Outage begins; dedup not yet involved.**
- **Jul 13** — Windows dedup enabled on FS01 (5 days into the unnoticed outage); rewrites ~10 TB, purges all FS01 shadow copies.
- **Jul 15** — KOR discovers the outage manually, re-accepts the new certificate, restarts jobs. **8-day zero-backup gap ends.**
- **Jul 15–18** — the restarted reverse-incremental job cannot fit the dedup churn; rollback file balloons to 8.9 TB against 1.8 TB free.
- **Jul 18–19** — job stopped to avert disk-full; stop took ~20 h to unwind; force-stopped per Veeam KB1727.
- **Jul 19–20** — FS01 re-protected on the Synology via forward incremental. **Resolved.**

Two distinct root causes: (1) an expired certificate + no monitoring = the 8-day outage (provider-side); (2) a dedup rollout executed without assessing backup ramifications = shadow-copy loss and the disk-full cascade (change-control side). The Jul 8–13 window is exposed by their intersection.

A1 — vCenter certificate expired: all backups dead ~8 DAYS FIXED 07-15

- Last good restore points **2026-07-07 ~17:00**; next are 07-15 (manual restart by KOR). **Zero backups for ~8 days.**
- Replacement certificate auto-issued 07-15 11:09 (valid to 2028-07-14, thumbprint `CD3550B8…4EA76E21`). Veeam pins the thumbprint → all jobs failed until manual re-accept.
- **No alert reached KOR at any point in the 8 days.** Found only because KOR opened the console.
- **Permanent exposure:** backups dead from 07-07 + shadow copies purged 07-13 ⇒ FS01 data created 07-08 → 07-13 and deleted before 07-15 has no recoverable copy.

A2 — Veeam B&R three years unpatched CRITICAL

- **12.0.0.1420** (v12 GA, early 2023) — predates multiple critical, actively-exploited Veeam CVEs. The backup server is the highest-value ransomware target in the estate.
- Cannot manage Windows Server 2025 machines (KOR-FS01, KOR-APP01). Remediation: in-place upgrade to **12.3.x** (rental licence via T-Net includes upgrades).

A3 — File-level restore of FS01 is impossible CRITICAL

- All FS01 restore points after 07-13 contain a Server 2025 **deduplicated** volume. Mount server KOR-BK01 is **Server 2019** and cannot mount that format — single-file restore fails. Full-VM and volume restores unaffected.
- Fix: after A2, add **KOR-APP01** (Server 2025, dedup feature installed 07-15) as managed server + repository mount server. Pre-07-13 points restore normally today.

A4 — Retired servers still in the nightly job WASTE

- **BMZ-HO-APP003 / APP004** decommissioned yet still objects in `Kor-VMs-New`, consuming window, repository space and copy bandwidth. (Removal in progress — item 21.)

A5 — Orphaned VM folders on the SAN RECLAIMABLE

- UC3200 (50.78 TB, 7.5 TB free) holds unregistered leftovers: `BMZ-HO-APP002`, `DC005`, `FS007`, `RDS001`.
- ⚠ `BMZ-HO-FS007` is the old BMZ file server — verify no unmigrated data before deletion; these folders predate current backup jobs and may be the only copy.

A6 — Duplicate RDS01 folders LANDMINE

- `Kor-RDS01` and `Kor-RDS01_1` both exist on UC3200; one holds the live server's disks. Verify via Edit Settings before touching either. Related: Veeam logged "*Kor-RDS01 is no longer processed by this job*" — confirm restore-point continuity.

A7 — FS01 shadow copies purged by the dedup rollout HISTORY LOSS

- The 07-13 optimisation rewrote ~10 TB and flushed all pre-existing VSS snapshots on E: — Previous Versions restarted from zero, uncommunicated. Older file versions remain recoverable from pre-07-13 Veeam points. Snapshots re-accumulating normally since.

A8 — Post-dedup catch-up run RESOLVED 07-20

- Root cause of the slowness: `Bottleneck: Target` at ~29 MB/s with transport already hot-add — the job ran **reverse incremental** (~3× write amplification per changed block). Repository disk + job mode, not SAN or network, was the limiter.
- The run could not fit and was stopped per KB1727. FS01 was then protected by a **forward-incremental job to the Synology**: 5.79 TB, ~126 MB/s, ~14 h, 0 errors.

A9 — Monitoring & governance gaps SYSTEMIC

- Jobs red for 8 days, zero alerts. Veeam e-mail notification config/recipients unknown — KOR receives nothing. Provider agents present on BK01 (Veeam Availability Console, NinjaRMM, ScreenConnect/TeamViewer) — none alerted.
- `ManualNew` job (4 VMs → Synology, unscheduled, holds a Mar-2026 FS01 full): purpose undocumented.
- C: was down to ~20 GB free on the backup server (cause: Administrator profile Downloads) — risked crashing Veeam's own database mid-run. Now reclaimed.

A10 — The E: repository is structurally over-committed NEW 07-20

File	Size	Status
`Kor-FS01…2026-07-15…vbk`	**14.7 TB**	**Chain head — load-bearing**
`Kor-FS01…2026-07-07…vrb`	**8.3 TB**	07-07 point, inflated by dedup churn — load-bearing
17 × `.vrb` (Jun 20 → Jul 7)	~87 GB	Daily history points — load-bearing
9 × `.vrb` dated Dec 2025	~44 GB	True orphans — no restore points reference them

- FS01's chain occupies **~23 TB of a 25.4 TB repository.** In reverse-incremental the `.vbk` holds the current state and all 18 older points roll backward from it — **nothing meaningful can be deleted without destroying the entire FS01 local history, including the pre-dedup Jul 1–7 points.** Only ~44 GB is safely reclaimable.
- **A design problem, not housekeeping:** a 21 TB VM was placed in a reverse-incremental chain on a repository that cannot hold one full plus rollback growth. The dedup rewrite merely exposed it.
- The 07-15 "Full" originates from the **failed** session; integrity **unverified** — validate before any decision about the chain.

A11 — Restore testing: first ever performed, 2026-07-20 DONE

No restore had ever been tested in this environment. Three were run on 2026-07-20; all restores were made to a new location (no originals touched).

Test	Result	Finding
1. FS01 file-level from the **local pre-dedup chain** (E:, Jul 1–7 point)	✅ **PASS**	The ~23 TB E: chain is genuinely restorable — its history is real, not merely catalogued.
2. FS01 file-level from the **off-site Cloud Connect copy** (T-Net DC, Jul 7 point)	✅ **PASS**	The off-site backstop works over the WAN. Two independently proven copies of FS01's pre-outage state.
3. FS01 file-level from the **new Synology backup** (post-dedup, Jul 20)	❌ **FAIL**	Veeam: "*Restoring files from deduplicated volumes (E:\) requires that the mount server … uses Windows Server 2012 R2 or later with data deduplication feature enabled.*"

- **Test 3 cause proven, cheap fix eliminated:** the Data Deduplication feature was installed on KOR-BK01 (Server 2019) and the server rebooted — **the error persisted.** This confirms a genuine format incompatibility: a Server 2019 mount server cannot read FS01's **Server 2025** dedup chunk store. A **Server 2025 mount server (KOR-APP01) is mandatory**, and Veeam must be upgraded to 12.3 first because 12.0 cannot manage a Server 2025 machine at all.
- **Current operational reality:** file-level restore of FS01 works for anything dated **before 2026-07-13** (from E: or the Cloud copy). For data created after that date the only path is Instant VM Recovery from the Synology backup, copying files out of the booted VM.

B. Questions for T-Net

1. Who monitors job results? Why did an **8-day total backup outage (07-07 → 07-15)** produce no alert to KOR? It was found only because KOR opened the console.
2. Who owns the vCenter certificate lifecycle? The expiry was foreseeable to the day, two years in advance.
3. What is the patch policy for backup infrastructure? B&R sat 3 years behind, across several exploited CVEs.
4. Why are retired BMZ-HO-* servers still in active jobs and on the SAN in 2026?
5. What restore-testing cadence is included in the engagement? When was the last successful test restore? (KOR can find no evidence of any — A11.)
6. **Design:** why is a 21 TB file server in **reverse-incremental** mode on a **25.4 TB** repository that cannot hold one full plus rollback growth? Forward incremental to the Synology did the same workload ~4× faster in ~⅓ the space. What design do you recommend?
7. **The E: chain (~23 TB):** validate and keep, or retire once Synology and Cloud copies are proven? What path preserves the pre-dedup Jul 1–7 history?
8. **Off-site for FS01:** on its own Synology chain it no longer flows to the Cloud Connect copy. What is the plan and timeline for the physical seed to the DC, and who performs it?
9. `ManualNew`**:** unscheduled, undocumented, targets the Synology (competing with FS01's new chain), holds a Mar-2026 FS01 full. Keep, repurpose or retire?
10. **Repository sizing:** what capacity do you recommend for E: (or its replacement) for a 21 TB and growing file server, and what is the refresh plan?
11. Request: add KOR (ilalonde@) to Veeam e-mail notifications regardless of provider monitoring, and document the alerting SLA.

C. Remediation punch list

#	Action	Owner	When	Verification
1	~~Re-accept vCenter certificate; restart jobs~~	KOR	**Done 07-15**	Jobs reconnected
2	~~Stop the doomed reverse-incremental run (KB1727); confirm snapshot removed~~	KOR	**Done 07-19**	vCenter confirmed snapshot deleted + consolidated
3	~~Reclaim C: on BK01; Windows updates + reboot; verify Veeam healthy~~	KOR	**Done 07-19**	49 GB free; services Running; repos rescanned
4	~~New `Kor-FS01` job → Synology, forward incremental, active full~~	KOR	**Done 07-20**	5.79 TB, ~14 h, 126 MB/s, 0 errors
5	Remove Kor-FS01 + BMZ-HO-APP003/004 from `Kor-VMs-New`; re-enable it, `Vcenter`, `Kor-Replication`	KOR	Now	3 small VMs nightly; off-site copy resumes

6	~~Test restores from all three copies (E: pre-dedup, Cloud Connect, Synology)~~	KOR	**Done 07-20**	2 PASS, 1 FAIL (cause proven) — see A11
7	Validate the unverified 07-15 "Full" (Veeam Backup Validator)	KOR / T-Net	Before any E: decision	Pass/fail recorded
8	Configure Veeam e-mail notifications to KOR; agree alerting SLA	T-Net	**Immediate**	Test alert received
9	Upgrade Veeam 12.0 → 12.3.x (config backup first)	KOR / T-Net	This week	Build ≥ 12.3; jobs green
10	Add KOR-APP01 as managed server + repository mount server (enables FS01 file-level restore)	KOR	After #9	FLR from a post-07-13 point succeeds
11	New backup-copy job for the `Kor-FS01` chain + physical seed to DC	T-Net / KOR	Scheduled	FS01 current in off-site copy
12	Decide the fate of the ~23 TB E: chain (A10) — only after #6 and #7	T-Net / KOR	After #6/#7	Documented decision
13	Switch `Kor-VMs-New` off reverse-incremental; review repository sizing	T-Net / KOR	With #9	Forward incremental; sizing documented
14	Delete ~44 GB orphaned Dec-2025 `.vrb` files (only safe filesystem reclaim)	KOR	Anytime	Files removed
15	Verify orphaned SAN folders; archive/delete (FS007 only after data review); resolve RDS01 / RDS01_1	KOR	After verification	UC3200 space reclaimed; single RDS01 folder
16	Clarify `ManualNew` purpose; document or retire	T-Net	Next review	Documented decision
17	Establish restore-test cadence (quarterly file-level, annual full-VM)	T-Net	Ongoing	Test log

D. Environment map (verified 2026-07-19/20)

- **Production:** vCenter 192.168.1.9; ESXi 192.168.1.10 + .16; cluster Van-Cluster; **UC3200 SAN** 50.78 TB (7.5 TB free). KOR-FS01 = 21 TB file server (Server 2025, Windows dedup on E: since 07-13).
- **Veeam head: KOR-BK01 — a VM** (Server 2019, workgroup, 16 GB, PostgreSQL config DB), B&R 12.0.0.1420, Enterprise Plus rental via T-Net.
- **Repositories:** `Default` → E:\Backup 25.4 TB (1.8 TB free, ~23 TB of it FS01's chain) · `Synology105` → F:\Backups 26.4 TB (5.79 TB free after FS01's new full) · `Kor-Replication` → **Cloud Connect at remote.t-net.ca, path \kor**, 25 TB (11.5 TB free).
- **Off-site:** Cloud Connect tenant hosted by T-Net; holds all 7 VMs incl. FS01, 8 points each, newest 07-07. Large fulls seeded by physically transporting hardware (WAN insufficient).
- **Jobs:** `Kor-VMs-New` (reverse-incr → E:) → `Vcenter` (reverse-incr → E:) → `Kor-Replication` (copy → Cloud), chained; `ManualNew` (→ Synology, unscheduled); **`Kor-FS01` (forward incr → Synology, new 07-20)**.

Compiled 2026-07-16, revised 2026-07-20, from live console, repository, filesystem and Veeam-database evidence. Findings verified at time of writing; configuration changes limited to those marked Done.